

NUMERICAL AND EXPERIMENTAL ANALYSIS OF HEAT TRANSFER IN AN ANNULAR MINICHANNEL USING STAR-CCM+

Magdalena Piasecka Norbert Dadas Beata Maciejewska Artur Piasecki

Kielce University of Technology, Al. Tysiąclecia Państwa Polskiego 7, 25-314 Kielce, Poland
 tmpmj@tu.kielce.pl

ABSTRACT

The present study combines experiments and CFD simulations to characterize heat transfer during HFE-649 flow in a vertical annular minichannel. The experiments were conducted at a fixed mass flow rate of 8 kg/h for two heater power inputs (Set #1 and Set #2), and the measured inlet fluid temperature, inlet/outlet pressures, and heater power input were used to define the boundary conditions. Based on the experimental conditions, Set #1 corresponds to single-phase flow, whereas Set #2 reaches the subcooled boiling region. A three-dimensional conjugate heat-transfer model was developed in Simcenter STAR-CCM+. For both sets, the simulations were performed using the Finite Volume Method (FVM). For Set #2, turbulence effects were represented using a Reynolds-Averaged Navier–Stokes (RANS) turbulence model. The CFD predictions were validated against measured axial wall-temperature profiles at thermocouple locations. Furthermore, local heat transfer coefficients were derived from the CFD fields using the wall heat flux and the wall-to-bulk temperature difference to quantify axial heat-transfer trends. The predicted temperature profiles remained in satisfactory agreement with the measurements for both sets. The resulting heat transfer coefficient distributions exhibit an entrance region with elevated values and downstream development that differs between Set #1 and Set #2, consistent with the higher heat input and the approach to subcooled boiling conditions in Set #2. The proposed modelling approach supports interpretation of the experimental data and provides a basis for further numerical studies of heat transfer in annular minichannels under controlled heating conditions.

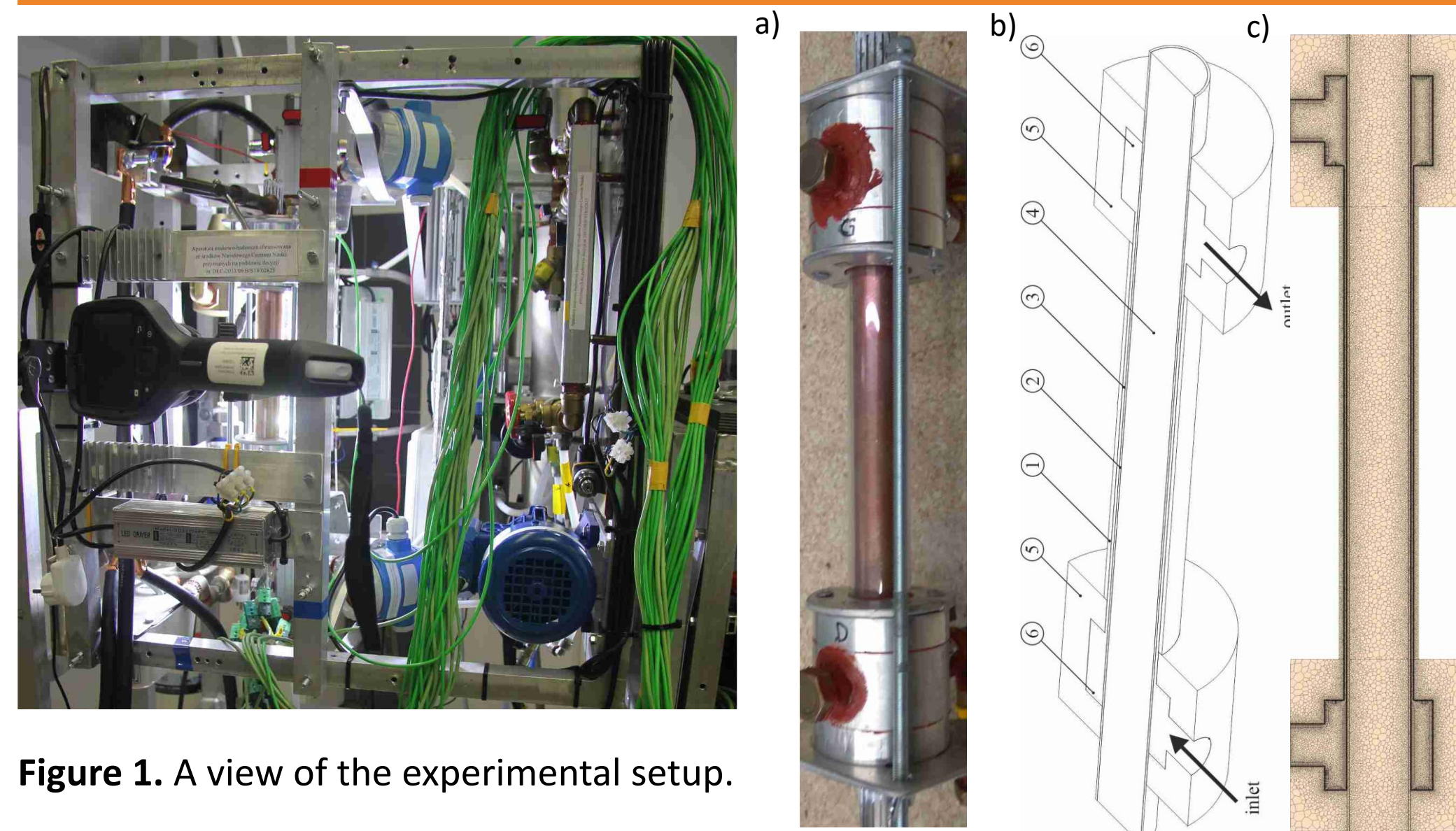


Figure 1. A view of the experimental setup.

A polyhedral volume mesh was generated for the entire computational domain using the automatic mesher in Simcenter STAR-CCM+. Near the inner and outer walls of the annular gap, several layers of prismatic cells were employed to resolve the steep gradients of velocity and temperature, with particular attention paid to the thin annular region where most of the thermal resistance is concentrated. Mesh statistics for Set #1 (lower heat input, single-phase flow) and Set #2 (higher heat input, subcooled boiling) are listed in Table 2. Grid convergence was assessed using Roache's Grid Convergence Index (GCI) method. The obtained GCI values indicated monotonic convergence and suggested that the solution is close to the asymptotic range.

Table 1. Operating conditions of the experimental series.

Main measured operating conditions	Set #1	Set #2
Mass flow rate (kg/s)	0.00222	0.00222
Fluid temperature at the inlet (K)	294.05	294.15
Fluid temperature at the outlet (K)	310.95	314.45
Gauge pressure at the inlet (kPa)	157	161
Gauge pressure at the outlet (kPa)	146	153
Heater power input (W)	72.8	91.8

Figure 2. The test module: (a) a view; (b) 3D digital model; 1 – glass tube, 2 – copper tube, 3 – annular minichannel, 4 – cartridge heater, 5 – module head with sealing, 6 – fluid nozzle; (c) the polyhedral mesh.

Table 2. Mesh statistics.

Element	Minichannel	Fluid inlet nozzle	Fluid outlet nozzle	Glass tube	Copper tube	Cartridge heater
Set #1						
cells	8 580 612	612 922	612 009	620 563	822 736	440 976
faces	38 981 486	4 097 499	4 089 432	3 854 012	4 730 921	2 904 814
verts	24 881 625	3 657 268	3 650 166	3 528 955	4 469 910	2 606 122
Set #2						
cells	10 271 773	789 900	800 115	665 726	1 831 398	388 124
faces	49 260 646	5 320 999	5 386 664	4 132 227	11 388 757	2 515 416
verts	33 498 411	4 738 350	4 797 819	3 787 352	10 431 574	2 269 572

Measured conditions of the experimental series, used as boundary conditions in STAR-CCM+, are listed in Table 1. The computational domain comprises:

- the fluid region representing the minichannel,
- the inner copper tube,
- the stainless-steel sheath of the cartridge heater, together with selected internal components required to capture axial heat conduction inside the heater,
- the outer borosilicate glass tube,
- the aluminum inlet and outlet nozzles for the fluid.

RESULTS AND OBSERVATIONS

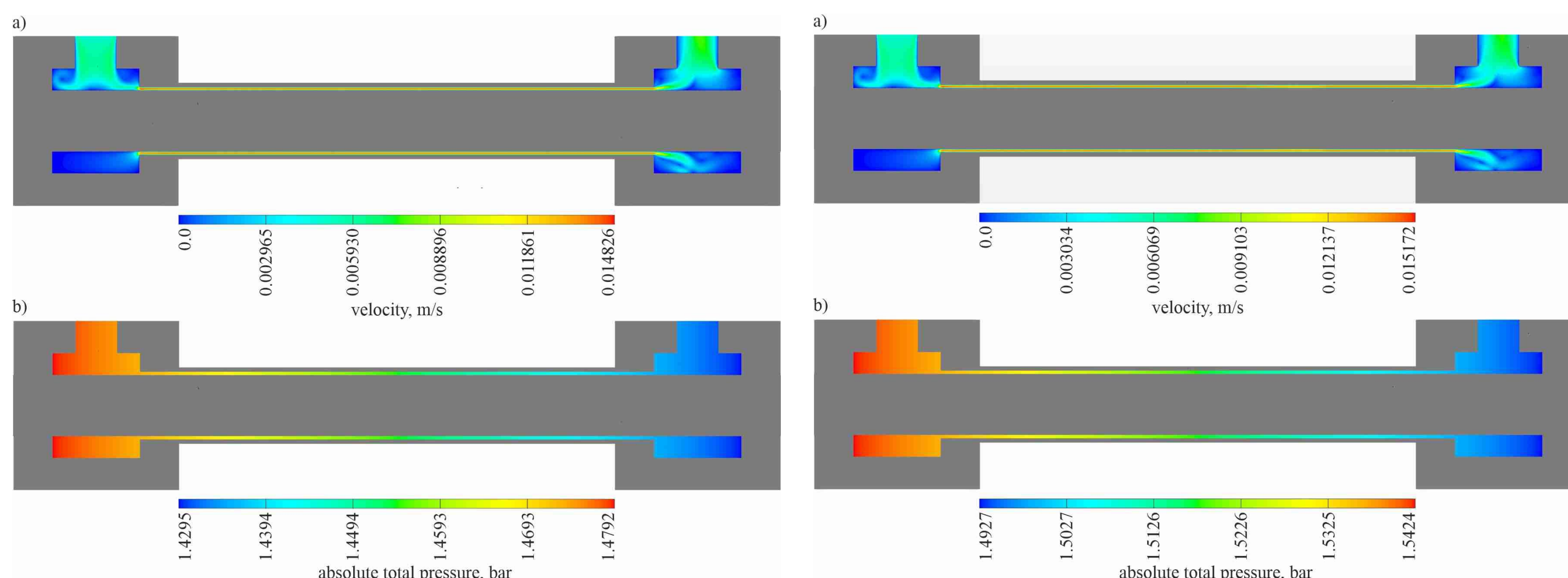


Figure 3. Distributions of velocity magnitude (a) and absolute pressure (b) in the fluid domain, Set #1.

Figure 4. Distributions of velocity magnitude (a) and absolute pressure (b) in the fluid domain, Set #2.

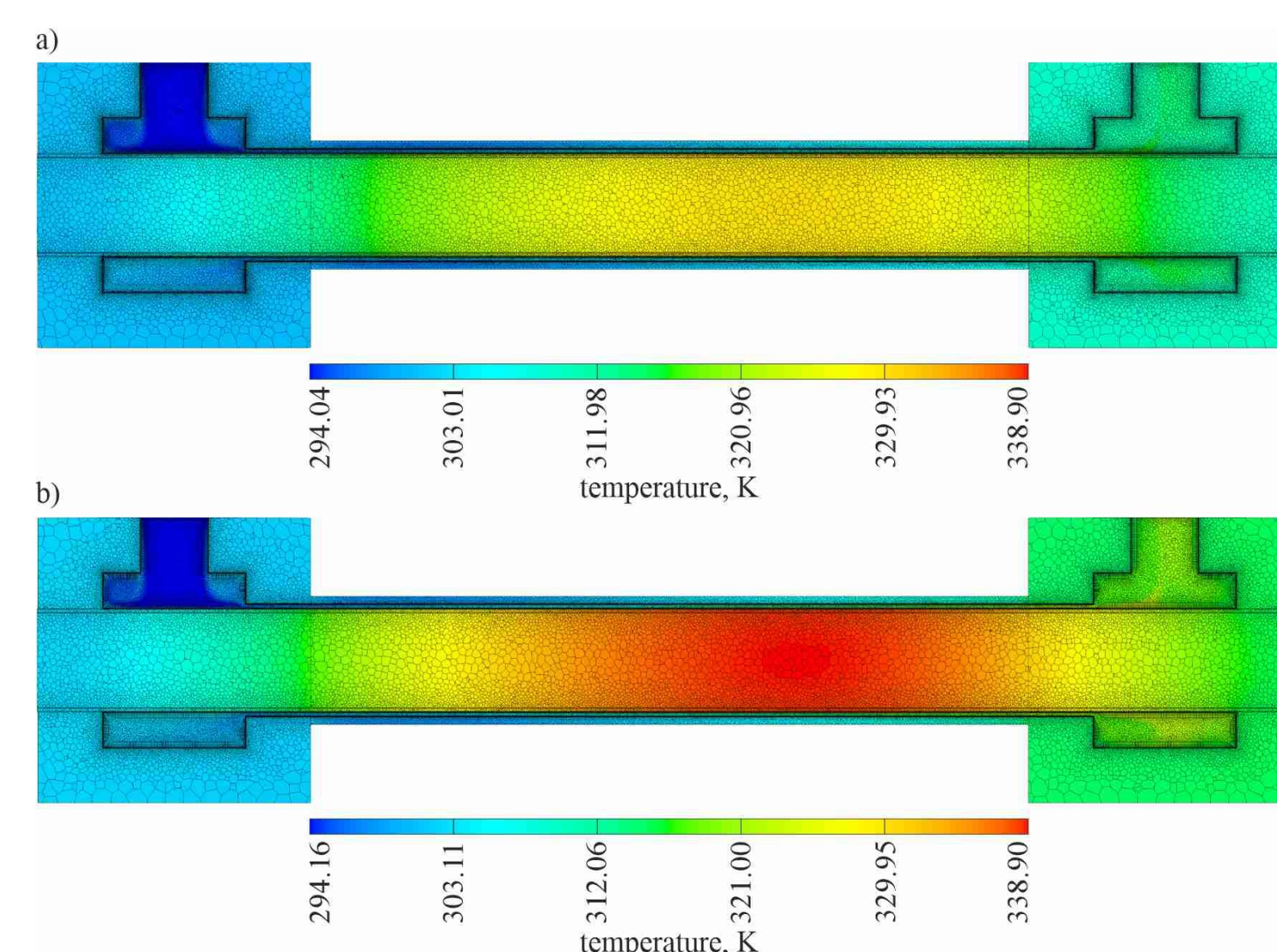


Figure 5. The temperature distribution in the entire test module as the results of CFD calculations in Simcenter STAR-CCM+, obtained for: a) Set #1, b) Set #2.

Table 3. Relative differences between the results obtained from numerical simulations in Simcenter STAR-CCM+ and the experimental data.

Set	Set #1	Set #2
Min relative difference δ_T (%)	0.001	0.24
Max relative difference δ_T (%)	3.58	3.82
Average relative difference δ_T (%)	1.66	2.05

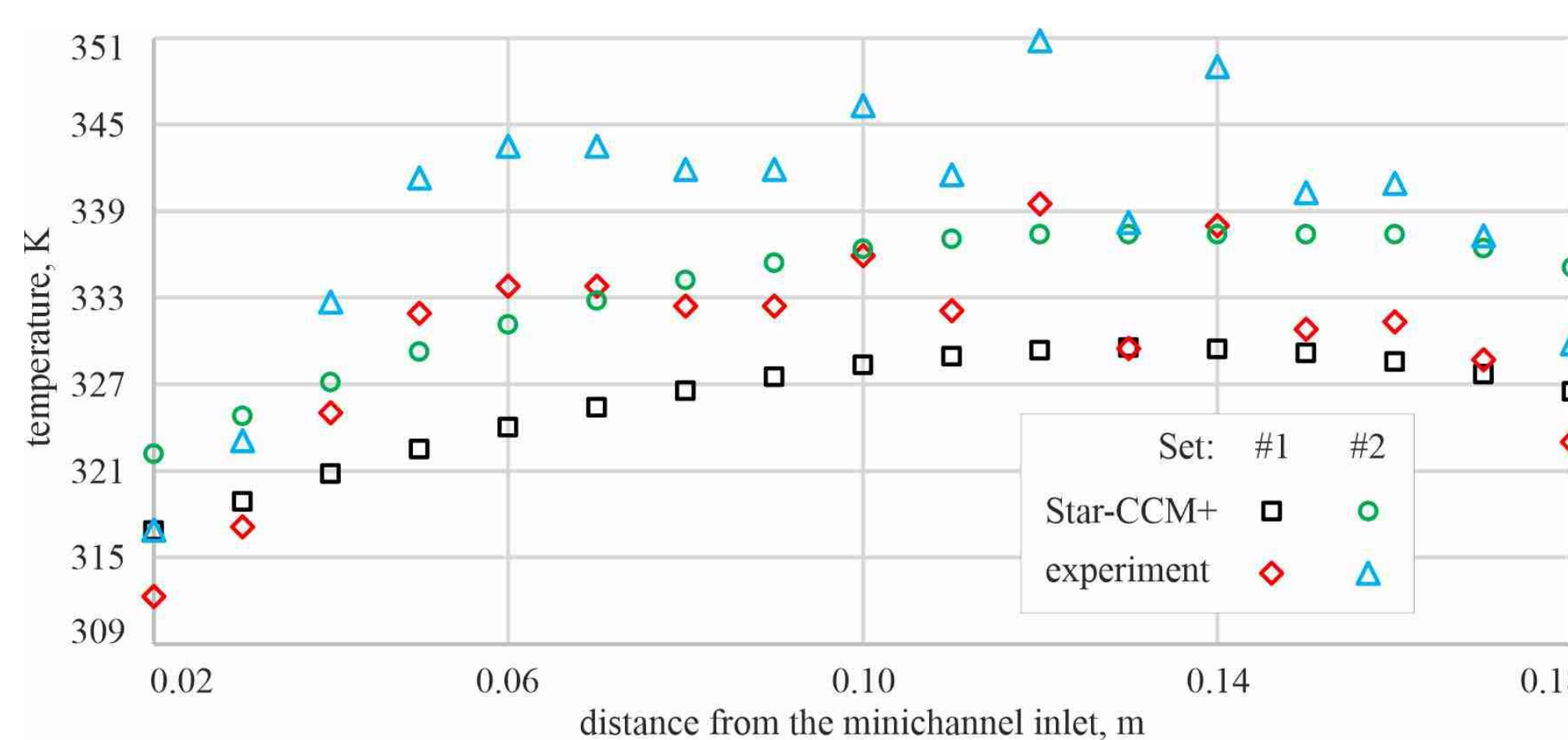


Figure 6. Axial wall-temperature profiles at the thermocouple locations: experimental and CFD results for Set #1 and Set #2.

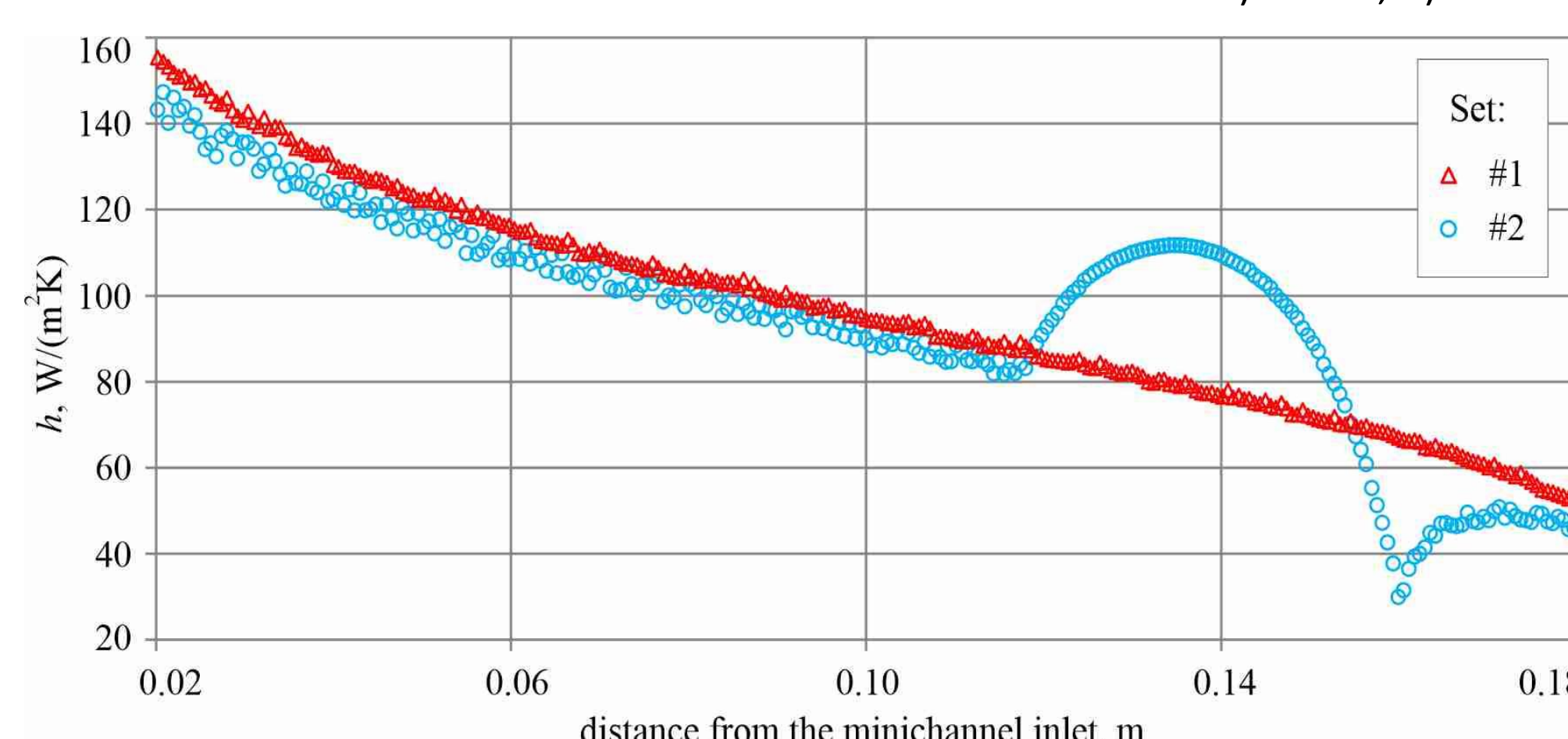


Figure 7. Axial distributions of heat transfer coefficient predicted by Simcenter STAR-CCM+ for Set #1 and Set #2.

- * For both heater power inputs, the heat transfer coefficient is relatively high in the entrance region.
- * For Set #1, the heat transfer coefficient decreases monotonically along the channel and reaches its minimum near the outlet.
- * For Set #2, the initial decrease in the heat transfer coefficient is followed, at approximately two-thirds of the heated length, by a local increase and a sharp drop. Further downstream, the heat transfer coefficient partially recovers and stabilizes near the outlet.
- * The local increase and subsequent drop observed for Set #2 are consistent with the onset of subcooled nucleate boiling.

CONCLUSIONS

- A three-dimensional conjugate heat-transfer (CHT) model was developed in Simcenter STAR-CCM+ for HFE-649 flow in a vertical annular minichannel and validated against experiments conducted at a fixed mass flow rate of 8 kg/h for two heater power inputs. The simulations used only measured operating conditions as boundary conditions; measured wall temperatures were not imposed.
- For both sets, the CFD results reproduced the overall shape and axial trend of the measured wall-temperature profiles. Outside the inlet and outlet developing regions, the predicted temperatures were generally lower than the experimental values, likely due to heat losses and measurement and modelling uncertainties. The average relative differences were 1.66% for Set #1 and 2.05% for Set #2.
- The CFD-derived heat transfer coefficient exhibited a pronounced entrance region. For Set #1, it decreased monotonically toward the outlet. For Set #2, the initial decrease was followed by a local increase and subsequent drop, after which the coefficient partially recovered and stabilized. This behavior is consistent with the onset of subcooled nucleate boiling.
- Overall, the experimental–numerical approach provides a reliable basis for interpreting heat-transfer behavior in the investigated annular minichannel and for further development of the numerical model.

ACKNOWLEDGMENTS: This research was funded by the National Science Centre, Poland, grant number UMO-2025/57/B/ST8/00907.